

# Extra Notes on Direct Limits

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Recall that for a directed system in  $\mathbf{Ab}$ , one can take its direct limit. However, if fixing a direct system  $I$ , and consider the functor category  $[I, \mathbf{Ab}]$  (sometimes even stronger,  $[I, \mathbf{CRing}]$ ), it has a natural abelian category structure:

**Proof:**

## 1. Zero Object:

If taking the „zero system“  $z : I \rightarrow \mathbf{Ab}$  by  $z(i) = 0$ , and  $zf_{ij} : z(i) \rightarrow z(j)$  to be the zero morphism, then for any other direct system  $F : I \rightarrow \mathbf{Ab}$ , the only possible natural transformation from  $z$ , say  $\mu : z \rightarrow F$ , is  $\mu(i) = 0 : z(i) = 0 \rightarrow F(i)$ ; and the only possible natural transformation to  $z$ , say  $\rho : F \rightarrow z$ , is  $\rho(i) = 0 : F(i) \rightarrow z(i) = 0$  (since  $0$  the trivial group is both initial and final in  $\mathbf{Ab}$ ).

Notice that this automatically makes  $\mu, \rho$  as natural transformations, as these two diagrams commute:

$$\begin{array}{ccc} z(i) & \xrightarrow{0} & z(j) \\ \mu_i=0 \downarrow & & \downarrow \mu_j=0 \\ F(i) & \xrightarrow{f_{ij}} & F(j) \\ & f_{ij} & \\ F(i) & \xrightarrow{f_{ij}} & F(j) \\ \rho_i=0 \downarrow & & \downarrow \rho_j=0 \\ z(i) & \xrightarrow{0} & z(j) \end{array}$$

So, we indeed have a zero object in this category.

## 2. Kernels and Cokernels:

If given a morphism  $\varphi : F \rightarrow G$  between two directed systems, for each  $i \in I$ , there exists a group homomorphism  $\varphi_i : F(i) \rightarrow G(i)$ , and hence one can define  $K(i)$  as the set-theoretic kernel of  $F(i)$ , and the morphism  $\ker(\varphi)_i : K(i) \hookrightarrow F(i)$  as the inclusion.

for the morphism  $f_{ij} : F(i) \rightarrow F(j)$  and  $g_{ij} : G(i) \rightarrow G(j)$ , notice we have the following naturality square, regarding the kernels:

$$\begin{array}{ccccc} K(i) & \xrightarrow{\ker(\varphi)_i} & F(i) & \xrightarrow{\varphi_i} & G(i) \\ & & \downarrow f_{ij} & & \downarrow g_{ij} \\ K(j) & \xrightarrow{\ker(\varphi)_j} & F(j) & \xrightarrow{\varphi_j} & G(j) \end{array}$$

Which, notice that one has  $0 = g_{ij} \circ (\varphi_i \circ \ker(\varphi)_i) = \varphi_j \circ (f_{ij} \circ \ker(\varphi)_i)$ , so the morphism  $f_{ij} \circ \ker(\varphi)_i$  uniquely factors through  $\ker(\varphi)_j : K(j) \hookrightarrow F(j)$ , say some  $k_{ij} : K(i) \rightarrow K(j)$ :

$$\begin{array}{ccccc}
K(i) & \xrightarrow{\ker(\varphi)_i} & F(i) & \xrightarrow{\varphi_i} & G(i) \\
\downarrow k_{ij} & & \downarrow f_{ij} & & \downarrow g_{ij} \\
K(j) & \xrightarrow{\ker(\varphi)_j} & F(j) & \xrightarrow{\varphi_j} & G(j)
\end{array}$$

Note the universality of kernels make  $K : I \rightarrow \mathbf{F}$  by  $i \mapsto K(i)$  and  $K(i \rightarrow j) = k_{ij}$  as a functor (or a directed system over  $I$ ).

Now, we claim that  $K$  together with the morphism  $\ker(\varphi) : K \rightarrow F$  is indeed a kernel of  $\varphi : F \rightarrow G$ : Suppose some other morphism  $\psi : L \rightarrow F$  satisfies  $\varphi \circ \psi = 0$ , then each index  $i \in I$  satisfies  $\varphi_i \circ \psi_i : L(i) \rightarrow F(i) \rightarrow G(i)$  being 0. So, the map uniquely factors through  $\ker(\varphi)_i : K(i) \rightarrow F(i)$ , say as  $\bar{\psi}_i : L(i) \rightarrow K(i)$ , as follow:

$$\begin{array}{ccccccc}
& & K(i) & \xrightarrow{\ker(\varphi)_i} & F(i) & \xrightarrow{\varphi_i} & G(i) \\
& & \searrow k_{ij} & & \searrow f_{ij} & & \searrow g_{ij} \\
\exists! \bar{\psi}_i \uparrow & L(i) & & K(j) & \xrightarrow{\ker(\varphi)_j} & F(j) & \xrightarrow{\varphi_j} & G(j) \\
& \nearrow \psi_i & & \nearrow \psi_j & & & & \\
& & L(j) & & & & & \\
& & \nwarrow l_{ij} & & \nwarrow \bar{\psi}_j & & & 
\end{array}$$

Which, the square between  $L(i), L(j), K(i), K(j)$  is a naturality square, because of the following diagram chase:

$$\begin{aligned}
\ker(\varphi)_j \circ (k_{ij} \circ \bar{\psi}_i) &= f_{ij} \circ (\ker(\varphi)_i \circ \bar{\psi}_i) = f_{ij} \circ \psi_i \\
&= \psi_j \circ l_{ij} = \ker(\varphi)_j \circ (\bar{\psi}_j \circ l_{ij})
\end{aligned} \tag{0.1}$$

Finally, using the fact that  $\ker(\varphi)_j$  is a kernel in  $\mathbf{Ab}$  (hence a monomorphism), this concludes that  $k_{ij} \circ \bar{\psi}_i = \bar{\psi}_j \circ l_{ij}$ , hence it's a naturality square. With  $i, j$  being arbitrary, one concludes that  $\bar{\psi} : L \rightarrow K$  is a natural transformation (with each component being  $\bar{\psi}_i$ ).

The uniqueness of  $\bar{\psi}$  (as a morphism in  $[I, \mathbf{Ab}]$ ) follows the universality of each  $\ker(\varphi)_i$  as a kernel in  $\mathbf{Ab}$ , proving that  $\ker(\varphi) : K \rightarrow F$  is indeed a kernel in  $[I, \mathbf{Ab}]$ .

**Note:** For the cokernel case, just put the arrows on the other side, the proof works similarly.

### 3. Equivalence of Monomorphism and Component-Wise Monomorphism:

(**Note:** The dual notion is epimorphism and component-wise epimorphism, where the proof follows similarly.)

It's clear that if a morphism  $\varphi : F \rightarrow G$  is component-wise monomorphic, it must be a monomorphism in  $[I, \mathbf{Ab}]$  (just because the left cancellative property applies to each component will prove the equality globally).

Now, suppose  $\varphi : F \rightarrow G$  is a monomorphism in  $[I, \mathbf{Ab}]$  (i.e. given any two morphism  $\rho, \mu : L \rightarrow F$ , one has  $\varphi \circ \rho = \varphi \circ \mu$  implies  $\rho = \mu$ ), then we claim the kernel of each component is 0 (which is equivalent to being monomorphic in  $\mathbf{Ab}$ ). Note that part 2 above helped us realize the kernel of the morphism as component-wise kernel. So, if consider the zero morphism  $0 : K \rightarrow F$  and the kernel morphism  $\ker(\varphi) : K \rightarrow F$ , notice that  $\varphi \circ 0 = \varphi \circ \ker(\varphi)$ , which the left cancellative property implies  $0 = \ker(\varphi)$ , or each component has  $0 = \ker(\varphi)_i : K(i) \rightarrow F(i)$ , proving each  $\varphi_i : F(i) \rightarrow G(i)$  is a monomorphism.

### 4. Equivalence of Monomorphism and Kernel:

(**Note:** The dual notion is epimorphism and cokernel, the proof also follows similarly).

Since kernel of a morphism is a component-wise kernel (cf. part 2), and kernel is the same as monomorphism in  $\mathbf{Ab}$ , then part 3 guarantees the kernel morphism in  $[I, \mathbf{Ab}]$  to be a monomorphism (as it's component-wise monomorphic).

Conversely, given a monomorphism  $\varphi : F \hookrightarrow G$ , running through similar proof of part 2 guarantees it has a cokernel  $\text{coker}(\varphi) : G \rightarrow C$  (where it's taken as cokernel component-wise). Then, since for every component  $\varphi_i : F(i) \rightarrow G(i)$  is a kernel of  $\text{coker}(\varphi)_i : G(i) \rightarrow C(i)$  (in  $\mathbf{Ab}$ , a monomorphism is a kernel of its own cokernel), hence globally  $\varphi : F \hookrightarrow G$  is a kernel of  $\text{coker}(\varphi) : G \rightarrow C$ , proving that  $\varphi$  is a kernel in  $[I, \mathbf{Ab}]$ .

## 5. Existence of Finite Biproduct:

Given two direct systems  $F, G \in [I, \mathbf{Ab}]$ , for each index  $i \in I$ , define the „biproduct“  $(F \oplus G)(i) := F(i) \oplus G(i)$ , which it naturally associates with projections  $\pi_{F,i} : (F \oplus G)(i) \twoheadrightarrow F(i)$  and  $\pi_{G,i} : (F \oplus G)(i) \twoheadrightarrow G(i)$ , also with inclusions  $\iota_{F,i} : F(i) \hookrightarrow (F \oplus G)(i)$  and  $\iota_{G,i} : G(i) \hookrightarrow (F \oplus G)(i)$ .

Now, we claim for every  $i \rightarrow j$  in  $I$ , there exists a natural morphism  $h_{ij} : (F \oplus G)(i) \rightarrow (F \oplus G)(j)$ . Let's consider the following diagram:

$$\begin{array}{ccccc}
 F(i) & \xleftarrow{\pi_{F,i}} & (F \oplus G)(i) & \xrightarrow{\pi_{G,i}} & G(i) \\
 \downarrow f_{ij} & & \downarrow \exists! h_{ij} & & \downarrow g_{ij} \\
 F(j) & \xleftarrow{\pi_{F,j}} & (F \oplus G)(j) & \xrightarrow{\pi_{G,j}} & G(j)
 \end{array}$$

Where, the uniqueness of  $h_{ij}$  is given by the two maps  $f_{ij} \circ \pi_{F,i} : (F \oplus G)(i) \rightarrow F(j)$ ,  $g_{ij} \circ \pi_{G,i} : (F \oplus G)(i) \rightarrow G(j)$ , and the fact that  $(F \oplus G)(j)$  (with the two projections) is a product of  $F(j), G(j)$  in  $\mathbf{Ab}$ .

Then, the association  $F \oplus G : I \rightarrow \mathbf{Ab}$  by  $i \mapsto (F \oplus G)(i)$ , and  $F \oplus G(i \rightarrow j) = h_{ij}$  is a functor by the universality of product.

(Note: If using the inclusions as the proof, we'll still get the same morphism, as the biproduct property guarantees the  $h_{ij}$  to agree, regardless if we use projections or inclusions to derive it). Also, the „projections“ and „inclusions“ component wise clearly form a naturality square, showing  $\pi_F : F \oplus G \rightarrow F$  (resp.  $\pi_G$  for  $G$ ) and  $\iota_F : F \rightarrow F \oplus G$  (resp.  $\iota_G$  for  $G$ ) are all morphisms in  $[I, \mathbf{Ab}]$ .

Finally, we claim the universality of  $F \oplus G$  as a biproduct (which we'll prove the product part, as the coproduct part is with similar proof).

□